

Chaotic operators need not be frequently hypercyclic

Literature-status packet for arXiv:1005.1416

Status

This is a `literature_already_answered` packet. It records a later answer to one subquestion explicitly mentioned in De la Rosa–Frerick–Grivaux–Peris, *Frequent hypercyclicity, chaos, and unconditional Schauder decompositions*, arXiv:1005.1416 [1].

Source question

In the introduction, immediately after proving that complex separable Banach spaces with unconditional Schauder decompositions support operators which are both chaotic and frequently hypercyclic, the source states:

“it is still an open question to know whether every chaotic operator on a Banach space is automatically frequently hypercyclic.”

This is distinct from the broader characterization question in the source and from the later real-space question about frequent hypercyclicity on Gowers’ real space.

Supporting answer

Quentin Menet’s later paper *Linear chaos and frequent hypercyclicity*, arXiv:1410.7173 [2], explicitly formulates the same problem:

“Is every chaotic operator frequently hypercyclic?”

Theorem 1.2 of that paper constructs a chaotic operator T on ℓ^1 which is not U -frequently hypercyclic and therefore is not frequently hypercyclic. This gives a negative answer to the source subquestion.

Scope limitations

This is not a new result from this run. It is a later-literature answer recorded to prevent rediscovery work on this subquestion. The source’s broad Question 1.1, asking for a characterization of complex separable Banach spaces supporting frequently hypercyclic or chaotic operators, remains outside this packet. The source’s final real-space question about whether the real Gowers space X_G supports a frequently hypercyclic operator is also not settled here.

References

- [1] M. De la Rosa, L. Frerick, S. Grivaux, and A. Peris, *Frequent hypercyclicity, chaos, and unconditional Schauder decompositions*, arXiv:1005.1416.
- [2] Q. Menet, *Linear chaos and frequent hypercyclicity*, arXiv:1410.7173.